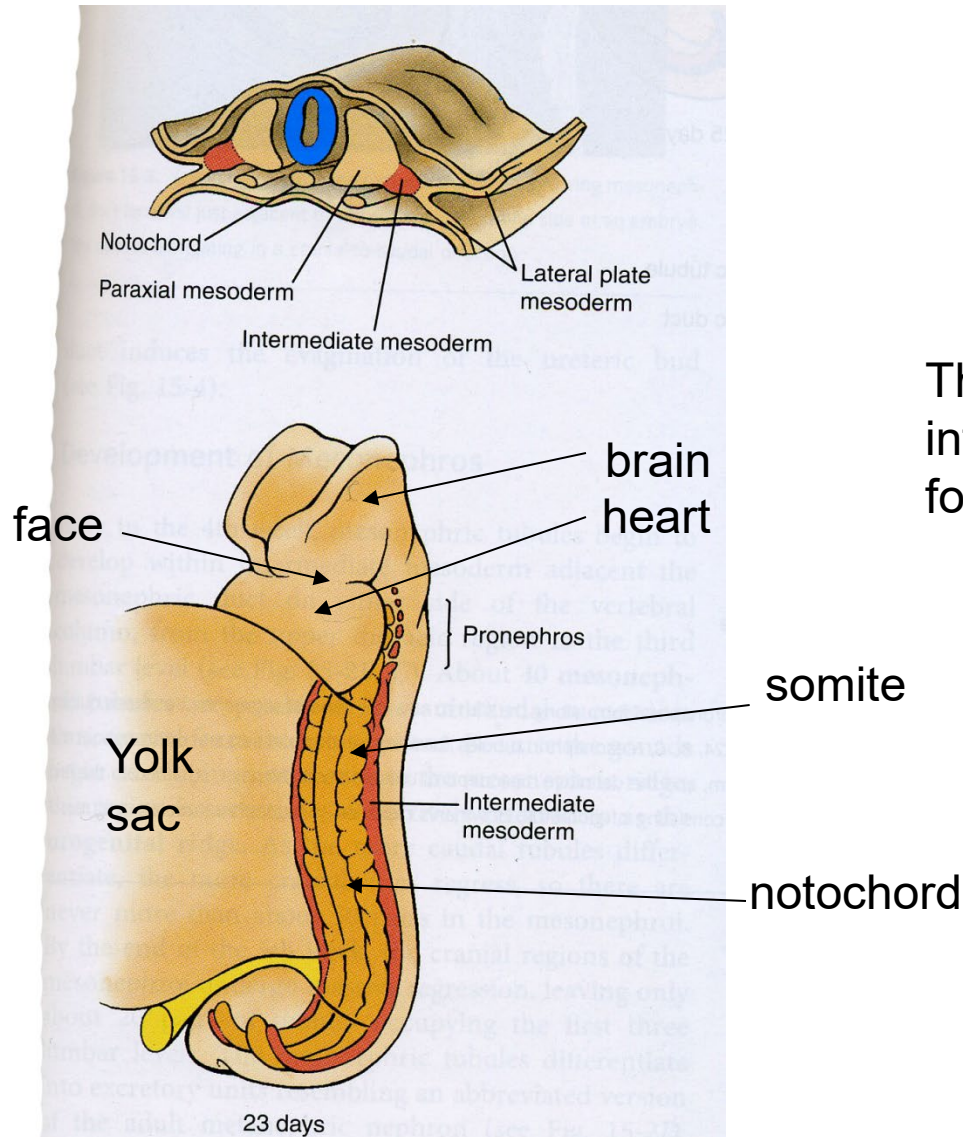


Session Objectives

1. embryology of the intermediate mesoderm and mesonephric duct systems
2. glomerulus
3. ureteric bud and metanephric blastema
4. embryologic derivatives of the cloaca and urogenital sinus
5. selected pathologies and embryologic anomalies provided



The intermediate mesoderm forms the kidneys

Figure 15-1. The intermediate mesoderm gives rise to paired, segmentally organized buds from the cervical to the sacral region. The pronephros is initially formed early in the 4th week in the cervical region.

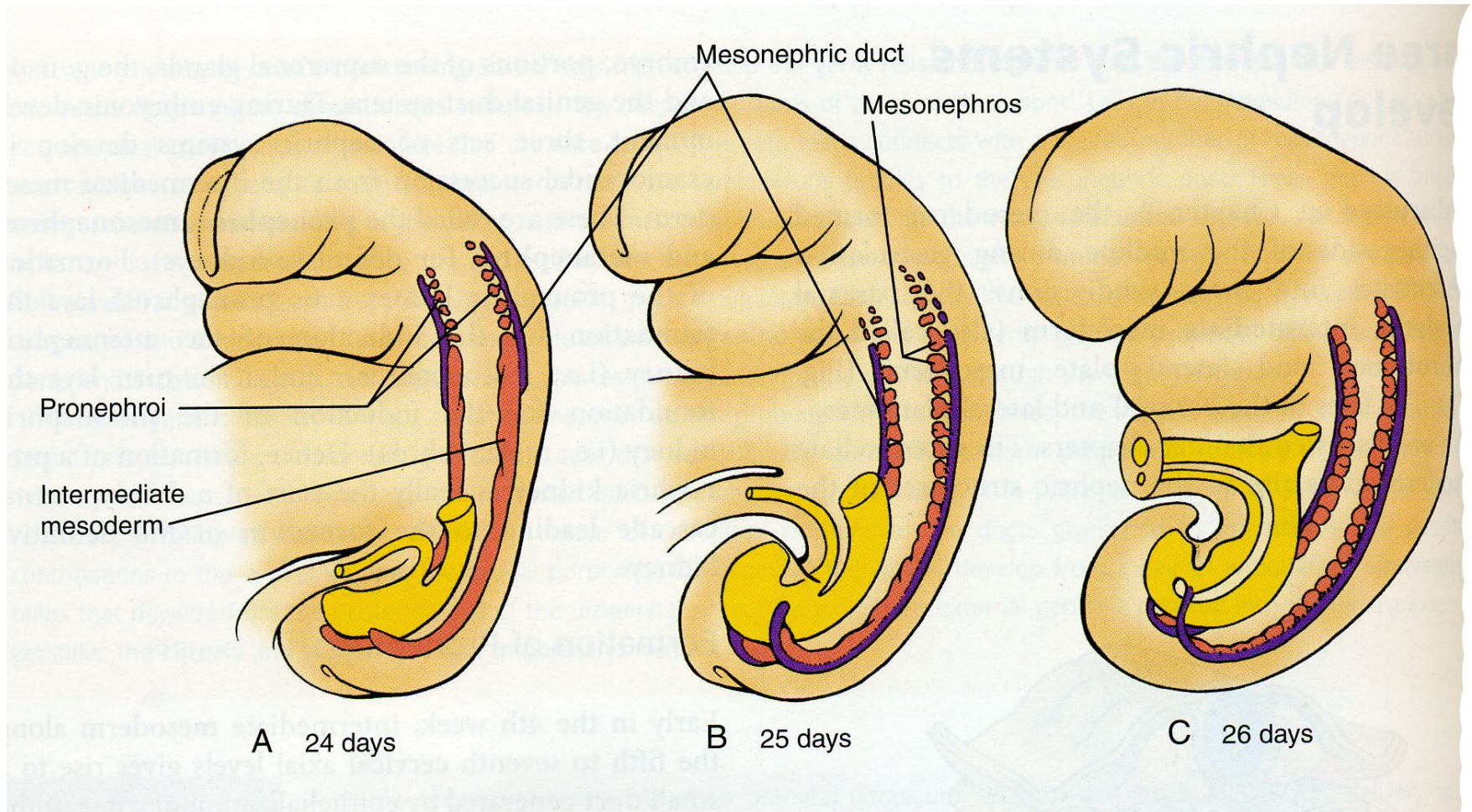
All nephric structures, derive from the intermediate mesoderm

Three types of kidneys found in embryos and also representing evolutionary stages and found in fishes

Pronephros is very primitive, and never develops into functional excretory structures in the human.

Mesonephros – its ducts and tubules remain in the adult – the glomeruli disappear in the adult.

Metanephros is the adult kidney with new glomeruli.



The mesonephros and the mesonephric kidney develops. This kidney disintegrates in part as the metanephric kidney forms. In part, as the mesonephric tubes remain.

Originally, in fishes the mesonephric kidney excreted urine and eggs and sperm. Later the mesonephric kidney got involved with male reproduction and the metanephric kidney took the role of urine excretion.

Wolffian duct = mesonephric duct

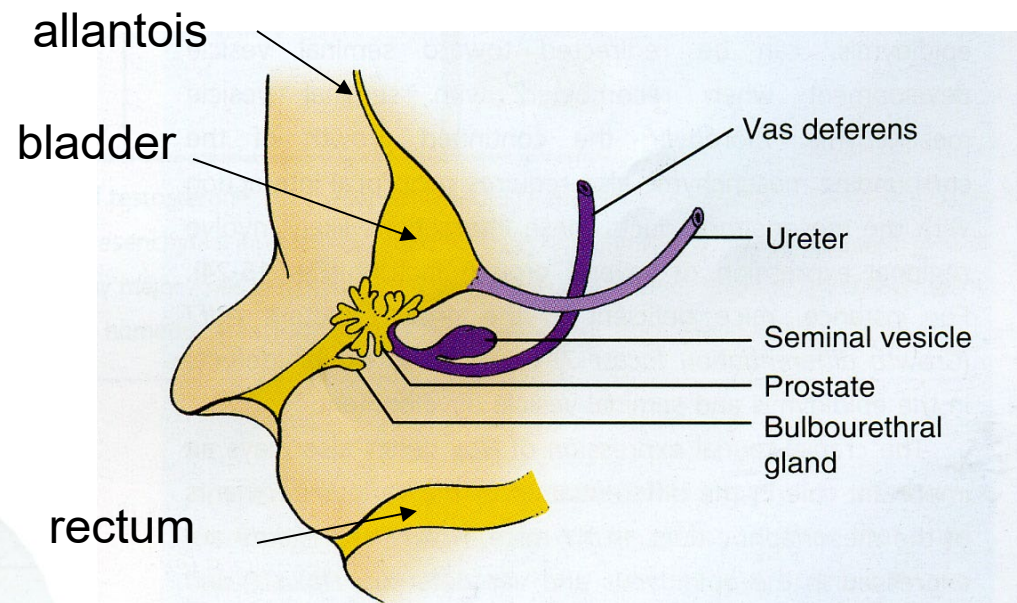
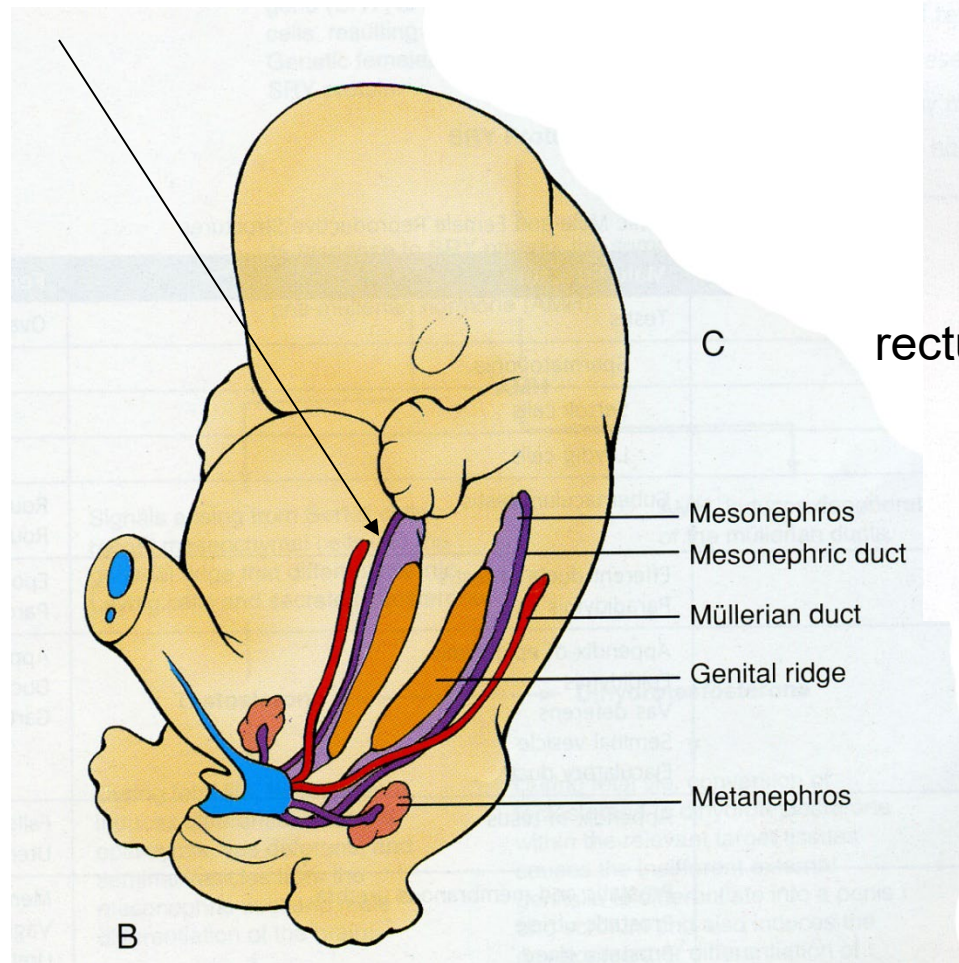


Figure 15-23. Development of the seminal vesicles, prostate, and bulbourethral glands. These glands are induced by androgens between the 10th and 12th weeks.

The colors in this slide are backwards

Mesonephric duct or Wolffian duct --- Müllerian duct = paramesonephric duct

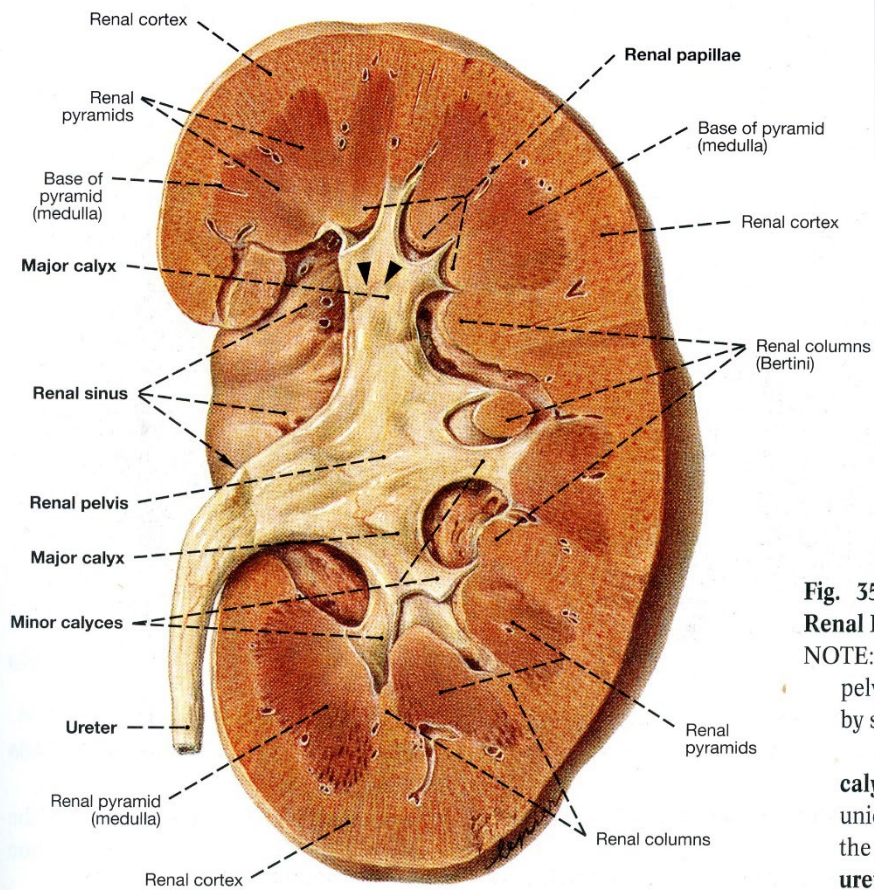
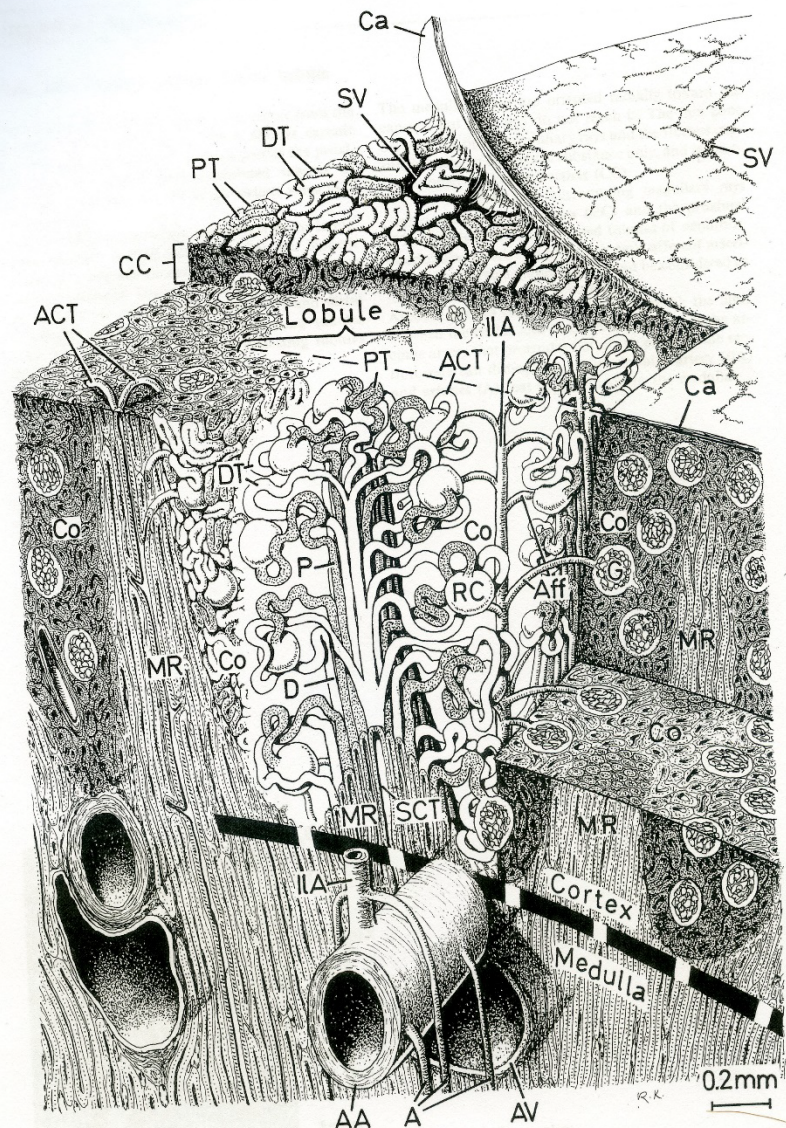
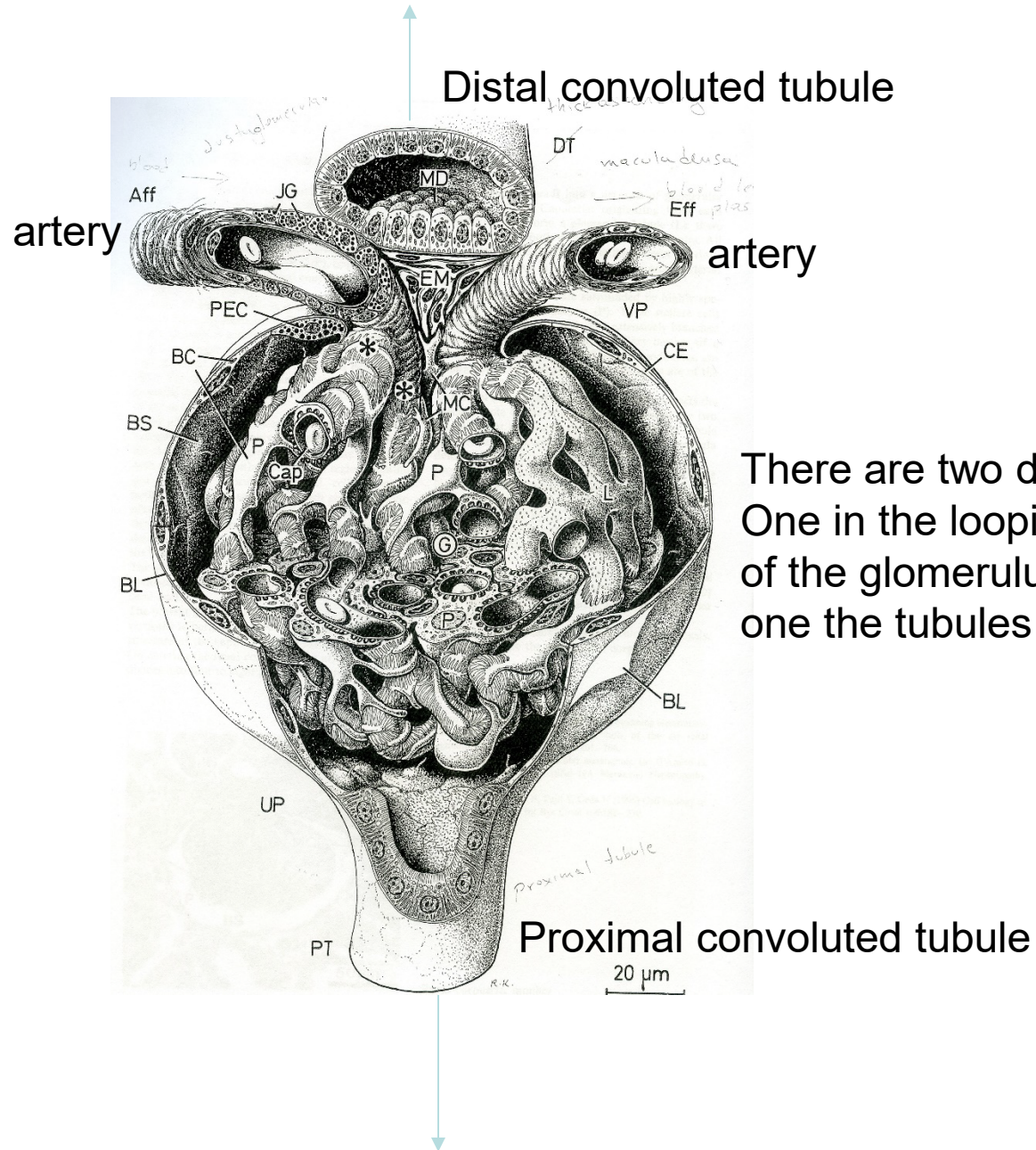


Fig. 32
Renal
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A glomerulus



Mesonephric - Ureteric bud forms the **collecting portion** of the kidney:
Ureter, renal pelvis, major and minor calyces, collecting ducts,
collecting tubules

Metanephric blastema sends signals to the ureteric bud causing it to
branch and grow, and the ureteric bud induces the blastema to
condense and form tubules INTERACTION

Metanephric blastema forms **excretory portion**: Bowman's capsule,
proximal convoluted tubule, loop of Henle, distal convoluted tubule

Metanephric kidney- ureteric buds branch from
the mesonephric ducts, and they interact with a mass of intermediate
mesoderm called the metanephric blastema
(also referred to as metanephric mesenchyme)

-

Fate of mesonephric kidney

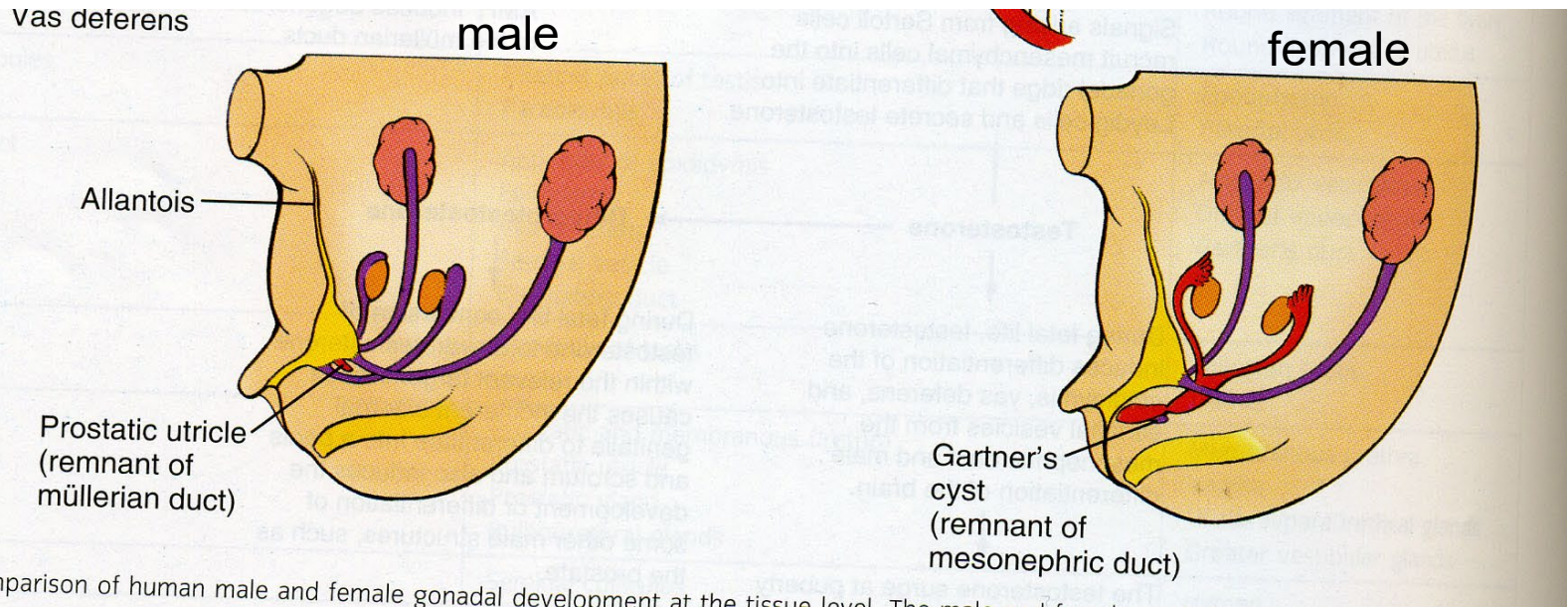
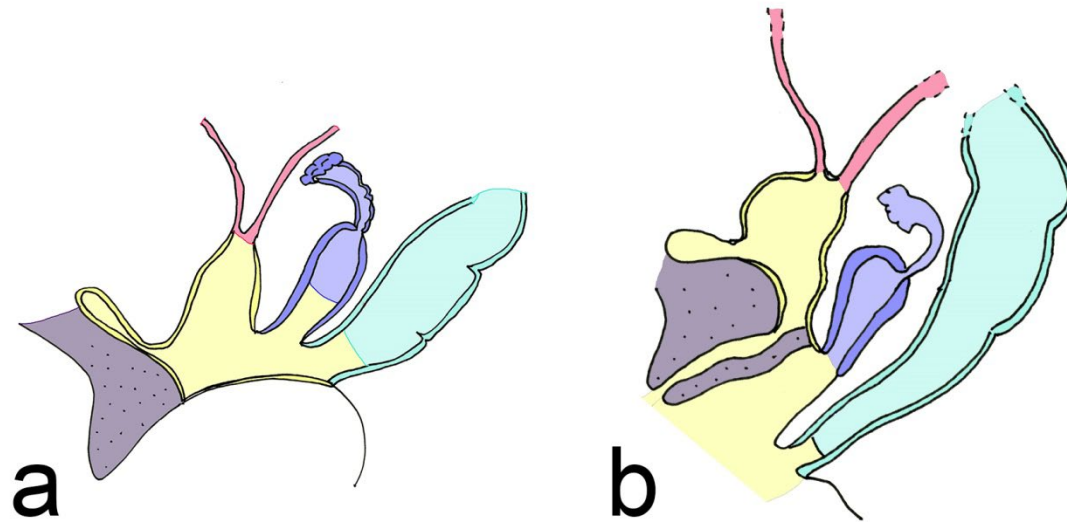


Figure 15-19. Comparison of human male and female gonadal development at the tissue level. The male and female reproductive systems are shown in cross-section.

Mesonephric kidney is not lost in adults –

Its tubing becomes incorporated in the reproductive system
of males – vas deferens and epididymis - trigone, ureter and metanephric calyces.

In females the paramesonephric duct is derived from the uterus as a diverticulum



The cloaca and its derivatives in a female. The cloaca is shown in yellow, genital tubercle in purple, paramesonephric duct in blue, and endoderm of the gastrointestinal tract in turquoise. A- Early stage, where the distal urinary, reproductive, and gastrointestinal tracts all have a common opening. The anterior out pouching towards the genital tubercle represents the allantois, which is an early drainage system for the fetus. B- Later stage where the urogenital system is separated from the distal GI tract via the urorectal septum. From anterior (left) to posterior (right), structures include the bladder and urethra, developing uterus and vagina, and anorectal canal. Watch the video.

The distal portion of the mesonephric ducts become incorporated into the posterior wall of the cloaca (process is called extrophy)- this forms the trigone of the bladder

Ureteric bud	Ureter, renal pelvis, major/minor calyces, collecting ducts, collecting tubules
Metanephric blastema	Bowman's capsule, proximal convoluted tubule, loop of henle, distal convoluted tubule
Mesonephric duct	Trigone of the bladder (and gives rise to the ureteric bud)
Urogenital sinus (part of the cloaca)	Bladder, urethra

Vascularization

Metanephric kidneys form low near the bladder and slowly ascend to the adult position

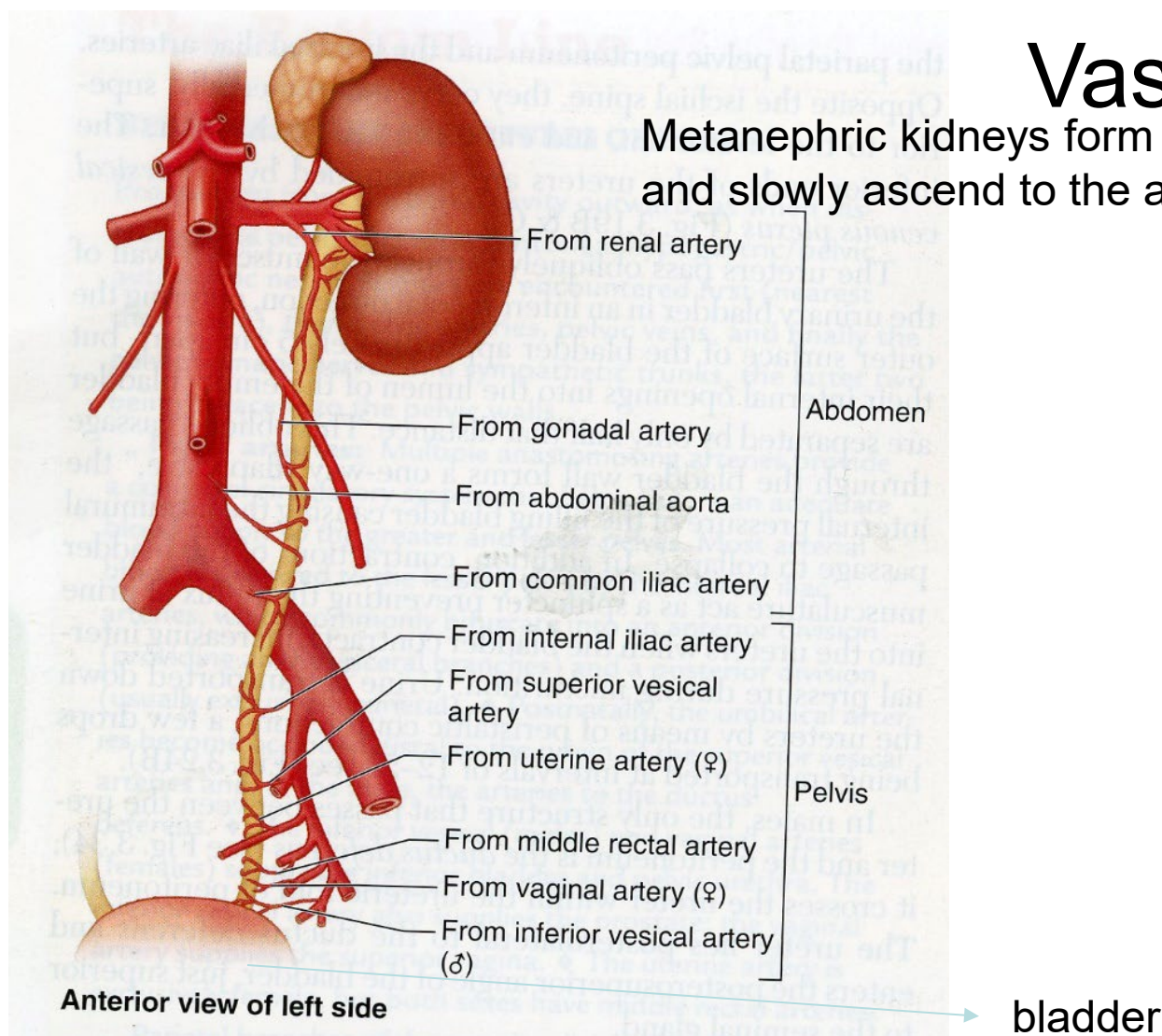
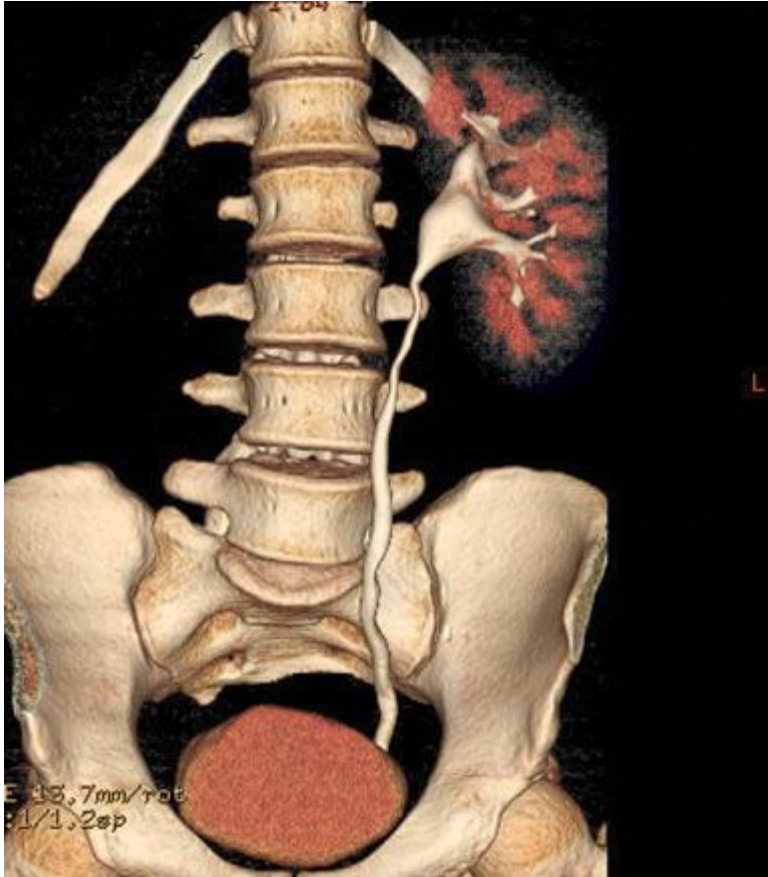


FIGURE 3.25. Blood supply of ureter. Branches supplying the abdominal half of the ureter approach medially, while those supplying the pelvic half approach laterally. During surgery, the ureters are avoided and left undisturbed when possible. If necessary, traction of the ureters is applied gently and only toward the blood supply to avoid disruption of the small branches.

Renal Agenesis



- Ureteric bud fails to contact the metanephric blastema → kidney does not form
- Unilateral: remaining kidney undergoes hypertrophy
- Bilateral: severe oligohydramnios, incompatible with life

Summary Slide

Ureteric bud	Ureter, renal pelvis, major/minor calyces, collecting ducts, collecting tubules
Metanephric blastema	Bowman's capsule, proximal convoluted tubule, loop of henle, distal convoluted tubule
Mesonephric duct	Trigone of the bladder (and gives rise to the ureteric bud) and vas deferens
Urogenital sinus (anterior part of the cloaca)	Bladder, pelvic urethra

-
- Oligo - means less
 - Poli – means much
 - Meso - means middle
 - Exstrophy - means inside out
 - Trigone – triangle
 - Agenesis – not growth – no development
 - Cloaca - Roman water clay tubes
 - Glomerulus – glomus ball of yarn (ulus is small)
 - Meta – after
 - Nephros - kidney